NMAP Scanning Assignment

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1. Shows all the available IP's available in the connected Network.

```
~ ❯ nmap -sn 172.20.184.0/22
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-11 21:58 +0530
Nmap scan report for 172.20.184.1
Host is up (0.0035s latency).
Nmap scan report for 172.20.184.26
Host is up (0.057s latency).
Nmap scan report for 172.20.184.27
Host is up (0.26s latency).
Nmap scan report for 172.20.184.28
Host is up (0.0044s latency).
Nmap scan report for 172.20.184.30
Host is up (0.040s latency).
Nmap scan report for 172.20.184.36
Host is up (0.012s latency).
Nmap scan report for 172.20.184.38
Host is up (0.0079s latency).
Nmap scan report for 172.20.184.46
Host is up (0.0066s latency).
Nmap scan report for 172.20.184.47
Host is up (0.059s latency).
Nmap scan report for 172.20.184.48
Host is up (0.0091s latency).
Nmap scan report for 172.20.184.52
Host is up (0.014s latency).
Nmap scan report for 172.20.184.53
Host is up (0.0057s latency).
Nmap scan report for 172.20.184.58
Host is up (0.45s latency).
Nmap scan report for 172.20.184.61
Host is up (0.014s latency).
Nmap scan report for 172.20.184.62
Host is up (0.017s latency).
Nmap scan report for 172.20.184.63
```

2. Scan the top 1000 ports of a given IP address.

```
~ ❯ nmap 172.20.186.48
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-11 22:00 +0530
Nmap scan report for 172.20.186.48
Host is up (0.0072s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT     STATE SERVICE
22/tcp   open  ssh
7070/tcp open  realserver

Nmap done: 1 IP address (1 host up) scanned in 0.69 seconds
```

3. Getting the services and OS Details of the given IP.

```
~ ❯ nmap -sV 172.20.186.48
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-11 22:07 +0530
Nmap scan report for 172.20.186.48
Host is up (0.0066s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT     STATE SERVICE         VERSION
22/tcp   open  ssh             OpenSSH 9.9p1 Ubuntu 3ubuntu3.2 (Ubuntu Linux; protocol 2.0)
7070/tcp open  ssl/realserver?
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.80 seconds
```

```
~ ❯ sudo nmap -sVO 172.20.186.48
[sudo] password for htarizzs:
WARNING: Disabling Service Scan (-sV) as it is incompatible with the IPProto Scan (-sO)
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-11 22:09 +0530
Nmap scan report for 172.20.186.48
Host is up (0.039s latency).
Not shown: 241 closed n/a protocols (proto-unreach)
PROTOCOL STATE         SERVICE
1        open          icmp
2        open|filtered igmp
6        open          tcp
17       open          udp
34       open|filtered 3pc
45       open|filtered idrp
51       open|filtered ah
58       open|filtered ipv6-icmp
80       open|filtered iso-ip
103      open|filtered pim
118      open|filtered stp
122      open|filtered sm
134      open|filtered rsvp-e2e-ignore
136      open|filtered udplite
216      open|filtered unknown
MAC Address: BC:F4:D4:75:52:55 (Cloud Network Technology Singapore PTE.)

Nmap done: 1 IP address (1 host up) scanned in 350.73 seconds
```

4. Scans multiple targets.

```
~ ❯ nmap 172.20.186.1-50
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-11 22:17 +0530
Nmap scan report for 172.20.186.2
Host is up (0.0047s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT     STATE SERVICE
5000/tcp open  upnp
7000/tcp open  afs3-fileserver

Nmap scan report for 172.20.186.6
Host is up (0.087s latency).
All 1000 scanned ports on 172.20.186.6 are in ignored states.
Not shown: 1000 closed tcp ports (conn-refused)

Nmap scan report for 172.20.186.9
Host is up (0.015s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds

Nmap scan report for 172.20.186.12
Host is up (0.023s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT    STATE SERVICE
111/tcp open  rpcbind

Nmap scan report for 172.20.186.16
Host is up (0.023s latency).
All 1000 scanned ports on 172.20.186.16 are in ignored states.
Not shown: 1000 closed tcp ports (conn-refused)

Nmap scan report for 172.20.186.29
Host is up (0.068s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE
49152/tcp open  unknown
49153/tcp open  unknown
62078/tcp open  iphone-sync

Nmap scan report for 172.20.186.36
Host is up (0.052s latency).
All 1000 scanned ports on 172.20.186.36 are in ignored states.
Not shown: 1000 closed tcp ports (conn-refused)

Nmap scan report for 172.20.186.37
Host is up (0.064s latency).
All 1000 scanned ports on 172.20.186.37 are in ignored states.
Not shown: 1000 closed tcp ports (conn-refused)
```